

Roschlynn Michael Dsouza

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SUMMARY

Multidisciplinary computer science graduate with hands-on experience in full-stack development, GenAI training, and applied ML research, proven by developing secure encryption systems (KP-ABE), P2P networks, and production-grade software. Collaborative, agile-focused developer committed to scalable and maintainable code.

EDUCATION

Saint Louis University

Master of Science in Computer Science

St. Louis, MO

Jan 2023 - Dec 2024

Don Bosco Institute of Technology

Bachelor of Engineering in Information Technology

Mumbai, India

Aug 2018 - Aug 2022

EXPERIENCE

Product Developer

Aidos, LLC - Ludus Lab

Feb 2025 – Present

St. Louis, MO

- Developed the "No Estimates Game," a multiplayer agile decision-making tool, using Node.js, Vue.js, React, and WebSockets for real-time interactions.
- Ensured code quality and automated delivery by writing comprehensive tests, implementing CircleCI pipelines, and collaborating with cross-functional teams in an Agile environment.

Front-End Engineer and GenAI/LLM Model Trainer

Outlier.AI - Remote

Mar 2025 – July 2025

St. Louis, MO

- Developed a React/Next.js frontend for an LLM training pipeline designed to generate modern website code.
- Ensured "golden response" quality for model fine-tuning by refining and rewriting LLM-generated code through a multi-turn supervised feedback process.

Research Assistant

Saint Louis University, School of Medicine

Oct 2024 – Feb 2025

St. Louis, MO

- Developed and validated ARIMA/SARIMA models to accurately forecast healthcare spending (2000-2040) and global energy trends (1973-2022).
- Conducted statistical analysis and presented data-driven visualizations to support research under Dr. Bahareh Rahmani.

Course Tutor & Teaching Assistant

Saint Louis University

Jan 2024 – Dec 2024

St. Louis, MO

- Evaluated assignments for over 65 students and mentored others in foundational CS courses, providing feedback to build academic confidence.

PUBLICATIONS

Energy Transitions over Five Decades: A Statistical Perspective on Global Energy Trends

Published in Computers (MDPI), May 13, 2025

DOI: 10.3390/computers14050190

- Analyzed global energy trends (1973-2022) using statistical analysis and SARIMA modeling to forecast growth in renewables and decline in fossil fuels.

Forecasting World Health Expenditures: A Hybrid AI and Generative AI Approach

Published in AIH (ACCSCIENCE Publishing), Sep 22, 2025

DOI: 10.36922/AIH025170033

- Enhanced healthcare expenditure forecasting by combining machine learning with generative AI to create synthetic data for LMICs.

KEY PROJECTS

P2P Chat Application | *Flask, IPFS, MySQL, Tailwind CSS, HTML, JS, TCP/IP Sockets* Aug 2024 – Dec 2024

- Designed a full-stack decentralized social network with real-time chat, group management, and distributed file sharing using IPFS and WebRTC.

KP-ABE Based OCR System | *Flask, OpenSSL, Charm-Crypto, Bash* Jan 2024 – May 2024

- Built a web service to extract text from images and share it securely using Key-Policy Attribute-Based Encryption (KP-ABE) over a distributed architecture.

Hand Sign Recognition using OpenCV | *Python, Tensorflow, OpenCV* Jan 2023 – Mar 2023

- Constructed a real-time gesture classification system using computer vision, achieving 85% accuracy in recognizing pre-defined hand signs.

TECHNICAL SKILLS

- **Languages:** C, C++, C#, Java, Python, JavaScript, Bash, Powershell, R
- **Web Development:** HTML5, CSS3, Tailwind CSS, Bootstrap, PHP, WordPress
- **Frontend Frameworks:** React, Vue.js
- **Backend & APIs:** Node.js, Express.js, Flask, Django, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Machine Learning / AI:** TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Seaborn
- **Cryptography & Security:** OpenSSL, Charm-Crypto, KP-ABE, TLS
- **Networking & Distributed Systems:** IPFS, WebRTC, TCP/IP Sockets
- **Cloud & DevOps:** AWS, Microsoft Azure, GCP, Docker, Linux, Windows Server, CircleCI